

A

Fixed spectrum; different interactions

- 105 perfect matchings M of $x_1 \dots x_8$

$$f_M = \frac{1}{2} \sum (i, j) \in M x_i x_j$$

- 35 four-plus-four partitions $A \mid A'$

$$f_A = \frac{1}{2} (\Pi_i \in A x_i + \Pi_i \notin A x_i)$$

- 24 nested-prefix controls (D–F)

 $\frac{1}{2} \times$ products of the first 2, 4, 6, 8 inputs of a seeded permutation

matching, quartic: input-gradient

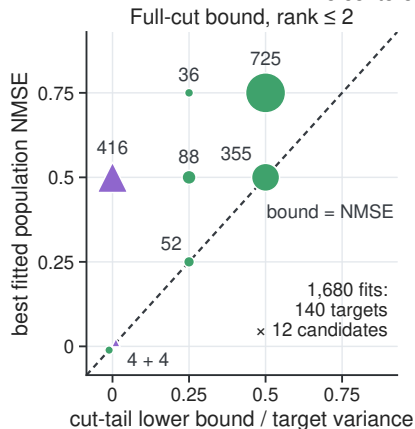
second moment $I_g / 4$, rank 8

nested: anisotropic spectrum

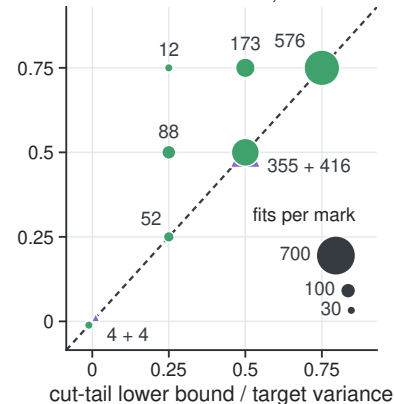
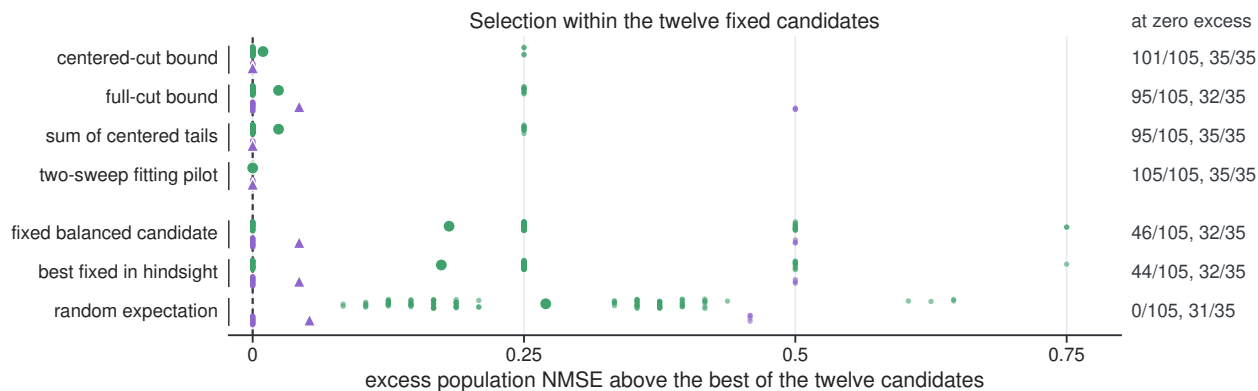
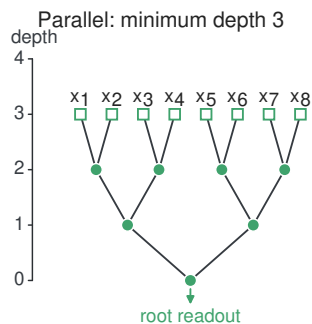
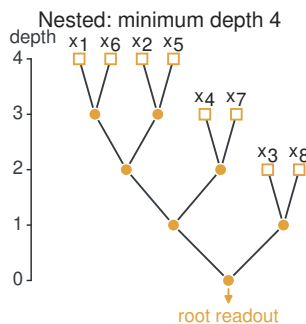
($\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, 1, 1$)

every tree: 7 multi-affine nodes,

28 coefficients, 14 edges, root readout

B

The centered constraint is stronger

Centered-cut bound, rank ≤ 1 **C****D****E****F**